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IQAC, DSATM

Department of Computer Science and Engineering

Case Study Report (CCA-2)

on

Impact of IIoT on Supply Chain and Warehouse Automation

Course Name: Industrial IOT (IIOT)

Course Code: BCS404

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1. Introduction

The Industrial Internet of Things (IIoT) has emerged as a cornerstone of Industry 4.0, interconnecting industrial machinery, sensors, and software through internet-enabled communication to enable real-time data collection and intelligent automation. In supply chain and warehouse operations, IIoT replaces error-prone manual workflows with smart, connected systems that improve inventory accuracy, operational efficiency, and customer satisfaction. Alherimi et al. ^[1] note in their IEEE Access systematic review that digital transformation of warehousing is revolutionizing operations by integrating technologies such as Automated Guided Vehicles (AGVs), IoT, robots, blockchain, and digital twins to address the increasing complexity of modern supply chains.

Globally, the industrial IoT market is projected to reach USD 503 billion by 2029 at a CAGR of 34.41%. This rapid growth reflects the scale of ongoing investment in IIoT-enabled industrial systems across sectors including logistics, manufacturing, and healthcare. This case study draws from four peer-reviewed papers published between 2024 and 2026 — two from IEEE journals/conferences and two from Scopus- and WoS-indexed journals — to analyze IIoT architecture, technologies, benefits, challenges, and adoption models for supply chain and warehouse automation.

2. Reference Papers

Ref	Title	Author(s)	Venue / Year	Index / DOI
[1]	Systematic Review of Optimization in Digital Warehousing Transformation	Alherimi, Saihi & Ben-Daya	IEEE Access, Sep 2024	IEEE / DOI: 10.1109/ACCESS.2024.10683736
[2]	Intelligent Warehousing: A ML and IoT Framework for Precision Inventory Optimization	Multiple Authors	IEEE Access, 2025	IEEE / DOI: 10.1109/ACCESS.2025.3614679
[3]	IoT-Based WMS Leveraging RFID and Cloud Platform Technologies	Multiple Authors	IEEE ATNT Conf., 2024	Scopus / DOI: 10.1109/ATNT61688.2024.10719252
[4]	IIoT & Industry 4.0 in Spare Parts Warehousing: A Systematic Literature Review	Khan, Solvang & Yu	Logistics, MDPI, 2024	Scopus & WoS / DOI: 10.3390/logistics8010016

Table 1: Reference Papers

3. IIoT Architecture for Warehouse Automation

A four-layer IIoT architecture governs the flow of data. The **Perception Layer** houses sensors, RFID readers, GPS modules, smart cameras, and AMRs that capture raw data. The **Network Layer** transmits data via Wi-Fi, 5G, LoRaWAN, and MQTT protocols through edge gateways. The **Processing Layer** applies cloud analytics, machine learning, and blockchain to transform raw data into intelligence. The **Application Layer** delivers insights via Warehouse Management Systems (WMS), ERP dashboards, and mobile apps.

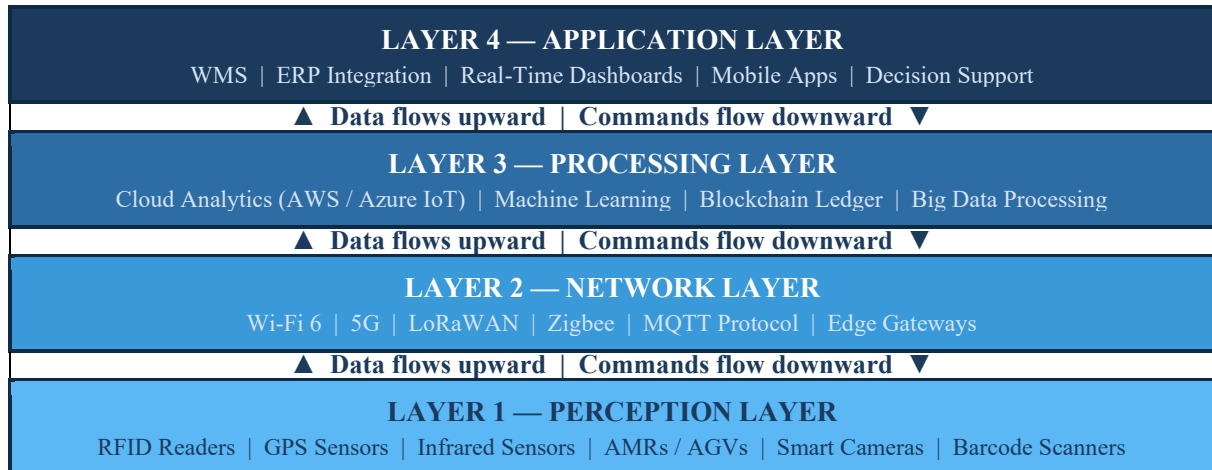


Figure 1: Four-Layer IIoT Architecture for Supply Chain and Warehouse Automation [1][4]

4. IIoT Technologies and Performance Impact

The research by Alherimi et al. ^[1] and Khan et al. ^[4] identify seven core IIoT technologies actively deployed in smart warehouses. Table 2 provides a comparative overview including function, application, and quantified benefit as reported in the reference papers.

Technology	Function	Warehouse Application	Reported Benefit
RFID	Wireless item identification	Inventory tracking, goods receipt	99.9% inventory accuracy [2]; eliminates manual scanning
AMRs / AGVs	Autonomous goods transport	Picking, put-away, replenishment	Transports 800+ items/hour; reduces labor costs by 30% [2]
Edge Computing	Low-latency data processing	Real-time rerouting of AMRs	Processes sensor data with <10 ms latency [2]
Blockchain	Immutable distributed ledger	End-to-end supply chain traceability	Tamper-proof records; enhances trust [1][4]
GPS / IoT Sensors	Real-time location tracking	Fleet and shipment monitoring	Proactive delay alerts; accurate ETA [3]
Cloud WMS	Scalable data management	Inventory, demand forecasting	Real-time synchronization; reduced discrepancies [3]
Environmental Sensors	Temp. / humidity monitoring	Cold-chain, perishables warehousing	Reduces perishable waste by 18% [2]

Table 2: IIoT Technologies and Quantified Benefits [1][2][3][4]

5. IIoT-Enabled Order Fulfillment Flowchart

Figure 2 presents the IIoT-driven order fulfillment process in a smart warehouse. The IIoT-Based WMS system described by the authors of reference [3] integrates RFID and Firebase Cloud to achieve real-time data synchronization at every stage. This replaces paper-based workflows with automated, sensor-driven operations, significantly reducing order processing time and human errors.

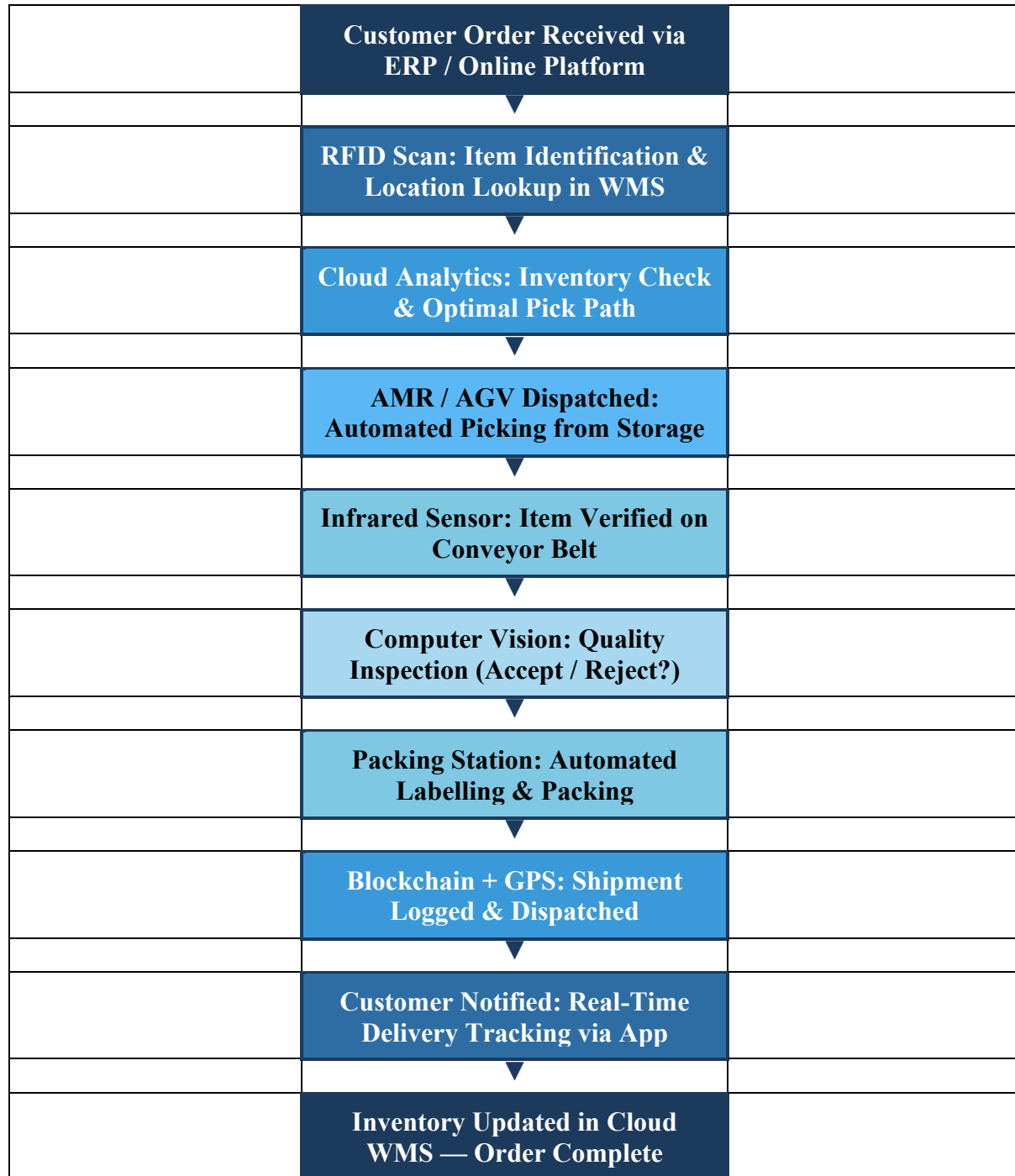


Figure 2: IIoT-Enabled Warehouse Order Fulfillment Process [1][3]

6. Benefits and Challenges

Figure 3 summarizes the key benefits and challenges of IIoT adoption in supply chain and warehouse automation, synthesized from all four reference papers ^{[1][2][3][4]}. Alherimi et al. ^[1] highlight that AGVs, IoT, and blockchain together address modern warehousing complexity, while Khan et al. ^[4] caution that in safety-critical sectors, implementation is often limited and controlled due to regulatory demands and high failure consequences.

✓ BENEFITS	✗ CHALLENGES
99.9% inventory accuracy via RFID [2]	High upfront infrastructure cost [1][4]
AMRs cut labor costs by 30% [2]	Cybersecurity risks from device connectivity [4]
<10 ms latency with edge computing [2]	Shortage of skilled IIoT professionals [4]
18% reduction in perishable waste [2]	Integration complexity with legacy ERP/WMS [1]
Blockchain ensures supply chain traceability [1]	Vendor lock-in with proprietary platforms [1]
Real-time WMS reduces order errors [3]	High ongoing maintenance costs [4]

Figure 3: Benefits vs. Challenges of IIoT in Supply Chain & Warehouse Automation [1][2][3][4]

7. IIoT Adoption Model

Based on the PRISMA-framework systematic review by Alherimi et al. ^[1] and the findings of Khan et al. ^[4], a six-phase adoption model is proposed for organizations integrating IIoT into warehouse operations. Figure 4 illustrates the model with a continuous risk mitigation layer spanning all phases.

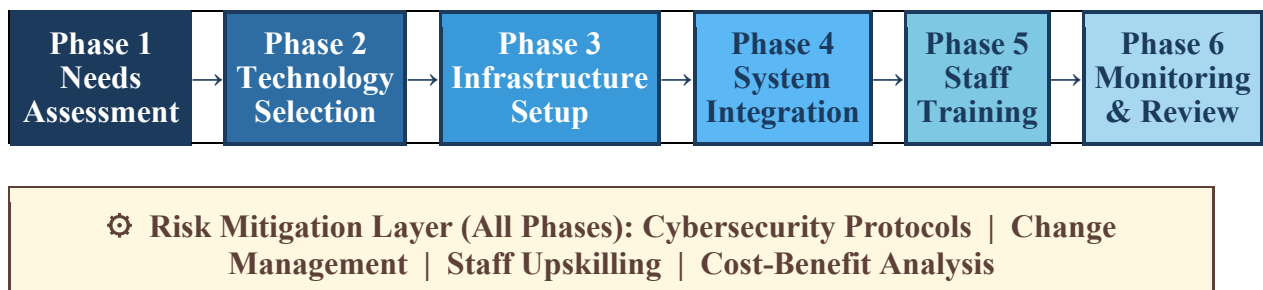


Figure 4: Six-Phase IIoT Adoption Model for Warehouse Management [1][4]

8. Conclusion

This case study demonstrates that IIoT integration in supply chain and warehouse automation delivers measurable improvements in inventory accuracy, operational efficiency, and customer responsiveness. Drawing from four peer-reviewed papers published between 2024 and 2026, the study shows that RFID-based systems achieve up to 99.9% inventory accuracy ^[2], AMRs reduce labor costs by 30% ^[2], and cloud-based WMS platforms enable real-time data synchronization and reduced discrepancies ^[3]. Alherimi et al. ^[1] confirm through a PRISMA-based systematic review that digital warehousing technologies including IoT, AGVs, blockchain, and digital twins are collectively reshaping warehouse operations at scale. Despite challenges including high implementation cost, cybersecurity risks, and integration complexity identified by Khan et al. ^[4], the structured six-phase adoption model provides a viable roadmap. As IIoT converges with AI, 5G, and blockchain, smart warehouses will increasingly operate autonomously, offering end-to-end supply chain transparency and adaptive logistics intelligence.

9. References

- [1] N. Alherimi, A. Saihi, and M. Ben-Daya, "A Systematic Review of Optimization Approaches Employed in Digital Warehousing Transformation," *IEEE Access*, vol. 12, Sep. 2024, doi: 10.1109/ACCESS.2024.10683736. **[IEEE]**
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- [3] A. A. Ali, R. A. Rashid, N. M. Abdikadir, A. A. Mohamed, and M. M. Ahmed, "IoT Based Warehouse Management System Leveraging On RFID and Cloud Platform Technologies," in *Proc. 2024 IEEE Int. Conf. on Advanced Telecommunication and Networking Technologies (ATNT)*, Sep. 2024, pp. 1–4, doi: 10.1109/ATNT61688.2024.10719252. **[IEEE / Scopus]**
- [4] N. Khan, W. D. Solvang, and H. Yu, "Industrial Internet of Things (IIoT) and Other Industry 4.0 Technologies in Spare Parts Warehousing in the Oil and Gas Industry: A Systematic Literature Review," *Logistics*, vol. 8, no. 1, p. 16, MDPI, Feb. 2024, doi: 10.3390/logistics8010016. **[Scopus & WoS]**